

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
GURUGRAM UNIVERSITY, GURUGRAM

Abstract

Air pollution is one of the most pressing environmental and public health challenges of the 21st century. According to the World Health Organization, more than 7 million people die annually due to air pollution-related diseases, and approximately 99% of the world's population breathes air that exceeds WHO quality guidelines. India's major cities consistently rank among the most polluted globally, with Delhi and Gurugram frequently recording Air Quality Index values in the Very Poor and Severe categories.

This project, Smart Air Quality Predictor, proposes a low-cost IoT-based solution that continuously monitors air quality parameters and forecasts future AQI trends using a lightweight machine learning model. The system is built around an ESP32 microcontroller connected to an MQ135 gas sensor for CO₂/VOC measurement and a DHT11 temperature and humidity sensor. Sensor readings are pushed to Firebase Realtime Database every 30 seconds through Wi-Fi.

A Firebase Cloud Function applies a linear regression model on recent sensor data to predict the next-hour AQI. The system features a real-time web dashboard built with Next.js and Chart.js, automated alerts when the predicted AQI exceeds unhealthy thresholds, and complete historical data logging. The total hardware cost is under INR 620, making it accessible for schools, homes, and research institutions.

The project demonstrates the integration of Internet of Things, cloud computing, and machine learning within a minimal budget, providing a practical alternative to expensive commercial AQI monitoring stations that cost several lakhs. The work contributes toward democratizing pollution intelligence at the community level.

STUDENT DECLARATION & SUPERVISOR APPROVAL
CERTIFICATE

We, Ishaan Jain and Sumit Kumar Pathak, bearing Roll Numbers 221011150016 and 221011150037 respectively, hereby declare that we have successfully prepared the project report titled Smart Air Quality Predictor.

We further declare that the project report has been completed by us and has been duly reviewed, finalized, and approved by our Supervisor/Guide. The work presented in this report is original and has not been submitted elsewhere for any other academic qualification.

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Student Signature: _____

Student Name: Sumit Kumar Pathak

Student Signature: _____

Supervisor/Guide Approval

I hereby certify that the above-mentioned project report has been reviewed and approved by me and is ready for submission.

Supervisor/Guide Signature: _____

Name of Supervisor: Dr. Jyoti Rani

Designation: Assistant Professor

Department: Bachelor of Technology

Institution: Gurugram University

Date: 30 April 2026

Place: Gurugram

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CERTIFICATE

This is to certify that the work contained in the PROJECT entitled Smart Air Quality Predictor, submitted by Ishaan Jain (221011150016) and Sumit Kumar Pathak (221011150037) for the award of the B.Tech. Computer Science and Engineering (IoT) to the Department of Engineering and Technology, Gurugram University, Gurugram, Haryana, is a record of bonafide work carried out by them under my direct supervision and guidance. I consider that the PROJECT has reached the standards and fulfills the requirements of the rules and regulations relating to the nature of the degree. The contents embodied in the PROJECT have not been submitted for the award of any other degree or diploma in this or any other university.

Date: 30 April 2026

Place: Gurugram

Supervisor Signature

Dr. Jyoti Rani

Assistant Professor

Bachelors of Technology

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
GURUGRAM UNIVERSITY GURUGRAM

CANDIDATE'S DECLARATION

I certify that

- a. The work contained in the project is original and has been done by us under the supervision of our supervisor.
- b. The work has not been submitted to any other Institute for any degree or diploma.
- c. We have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- d. Whenever we have used materials from other sources, we have given due credit by citing them in the text of the project and in the references.
- e. Whenever we have quoted written materials from other sources, due credit is given to the sources by citing them.
- f. From the plagiarism test, it is found that the similarity index of the whole project is within 25% and single paper is less than 10% as per the university guidelines.

Date: 30 April 2026

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Student Signatures

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Chapter 1: Introduction

Air pollution has emerged as one of the most critical and complex environmental challenges of the twenty-first century. Rapid industrialisation, exponential growth in vehicular traffic, large-scale construction activity, and seasonal agricultural burning have collectively driven atmospheric pollutant concentrations far beyond safe thresholds in cities across South Asia, East Asia, Sub-Saharan Africa, and Latin America. The World Health Organization estimates that more than 7 million premature deaths occur globally every year as a direct consequence of outdoor and household air pollution.

In the Indian context, the situation is particularly alarming. India is home to several of the world's most polluted cities, and Delhi, Gurugram, Noida, Ghaziabad, and Faridabad routinely breach severe AQI thresholds during the October-February winter season. Atmospheric temperature inversions trap vehicular exhaust, industrial emissions, and crop-residue smoke in a dense near-surface layer.

The Smart Air Quality Predictor project addresses these gaps by designing, building, and validating a low-cost, open-source IoT-based environmental monitoring system that integrates real-time sensing, cloud data pipelines, machine learning forecasting, and an interactive web dashboard within a hardware budget of under INR 620.

1.1 Background

The Air Quality Index communicates daily air quality information in an intuitive form. India's National AQI framework is administered by the Central Pollution Control Board and considers pollutants including PM_{2.5}, PM₁₀, NO₂, SO₂, CO, O₃, NH₃, and Pb. Each pollutant sub-index is computed using breakpoint concentration tables, and the overall AQI is the maximum sub-index across measured pollutants.

AQI Range	Category	Health Implication
0-50	Good	Minimal impact; suitable for all outdoor activities.
51-100	Satisfactory	Minor discomfort for sensitive individuals.
101-200	Moderate	Breathing discomfort for asthma and heart patients.
201-300	Poor	Breathing discomfort for most people.
301-400	Very Poor	Respiratory illness on prolonged exposure.
401+	Severe	Serious health risk; avoid outdoor exposure.

1.2 Motivation

The primary motivation for this project is rooted in lived experience. Both project members reside and study in Gurugram, a city that illustrates urban air pollution at its most extreme. During winter, vehicular exhaust, industrial emissions, construction dust, and residue smoke create a near-permanent haze.

Most residents have no real-time access to hyperlocal AQI data for their specific neighbourhood. Existing apps and government websites report city-level averages or readings from fixed monitoring stations that may be several kilometres away and unrepresentative of local conditions.

The project also aligns with the Government of India's National Clean Air Programme and the global movement toward citizen science and open environmental data. Affordable community-deployable sensor nodes can complement official networks by providing denser hyperlocal data.

1.3 Scope of the Project

- **Real-time continuous sensing** of CO₂ and VOC using the MQ135 gas sensor with temperature and humidity acquisition using DHT11.
- **Wireless transmission** of structured JSON sensor payloads to Firebase Realtime Database at a configurable interval.
- **Machine learning forecasting** of next-hour AQI using linear regression on recent historical readings.
- **Responsive web dashboard** built with Next.js and Chart.js for live readings, historical trends, and forecast overlays.
- **Automated alerting** through email and push notifications when predicted AQI crosses unhealthy thresholds.

1.4 Problem Overview

A comparative analysis of existing air quality monitoring solutions shows that no commercially available low-budget system provides real-time sensing, predictive intelligence, cloud storage, and automated alerting simultaneously. The Smart Air Quality Predictor uniquely combines these capabilities at an accessible cost.

System / Product	Cost	Real-Time	Predictive	Cloud	Alerts
CPCB CAAQMS Station	5-50 Lakhs	Yes	No	Yes	Limited
AirVisual Pro	20,000+	Yes	Limited	Yes	Yes
Basic Arduino + MQ135	800	Yes	No	No	No
Smart Air Quality Predictor	580-620	Yes	Yes	Yes	Yes

Chapter 2: Literature Review

A systematic review was conducted to establish the theoretical, scientific, and engineering foundations required for the Smart Air Quality Predictor system. The review focuses on low-cost sensing technology, robust IoT and serverless cloud architecture, and short-term time-series forecasting models.

2.1 Review Methodology and Foundational Concepts

The literature was sourced from peer-reviewed publications, environmental science databases, manufacturer datasheets, and platform documentation from Firebase, Espressif, and CPCB. The CPCB National AQI framework provides the project's primary output metric and health alerting basis.

2.2 Low-Cost Chemiresistive Sensing and Environmental Compensation

Research confirms that MQ-series sensors can serve as effective proxies when properly conditioned. Studies show that MQ135 sensors, after burn-in and calibration, can correlate strongly with CO₂ and VOC trends. Pairing MQ135 with DHT11 helps compensate for humidity-related cross-sensitivity.

2.3 Serverless IoT Architecture and Cloud Data Pipeline

The ESP32 provides integrated Wi-Fi, dual-core processing, and ADC capabilities in a low-cost package. Firebase Realtime Database and Cloud Functions support low-latency ingestion, persistent storage, real-time dashboard synchronization, and serverless model execution.

2.4 Time Series Forecasting and Modeling Rationale

Linear regression was selected because it is lightweight, interpretable, and suitable for short-term next-hour prediction. While LSTM and ARIMA models can capture more complex patterns, the simple model is adequate for distinguishing major AQI category transitions within the project's constraints.

2.5 Comparative Analysis and Gap Identification

The review identifies a gap between expensive regulatory systems and affordable hobbyist prototypes. Existing low-cost systems generally lack cloud persistence, automated alerts, and prediction. This project fills that feature-complete low-budget segment.

Chapter 3: System Analysis and Requirements

The proposed air quality monitoring system is designed as a low-cost IoT-enabled platform capable of real-time sensing, cloud integration, and predictive analysis. The architecture combines hardware sensors, wireless communication, cloud storage, and machine learning components into a unified framework.

O1: Real-Time Sensing

Continuously measure gas concentration using MQ135 and environmental conditions using DHT11. Readings are averaged to reduce ADC noise while preserving slowly varying pollution dynamics.

O2: Cloud Data Pipeline

Transmit structured sensor data from ESP32 to Firebase Realtime Database every 30 seconds with automatic error handling and retry logic.

O3: ML-Based Forecasting

Generate next-hour AQI predictions from recent historical data using ordinary least squares linear regression in a serverless Cloud Function.

O4: Alert System

Notify registered users through email and push notifications when predicted AQI exceeds the unhealthy threshold, while enforcing a cooldown to prevent notification fatigue.

O5: Web Dashboard

Provide real-time visualisation of live sensor data, historical trends, forecast values, temperature, humidity, and alert status through a responsive web dashboard.

O6: Low-Cost Deployment

Achieve the system within a total hardware cost under INR 620 using free and open-source software tools and free-tier cloud services.

Hardware Bill of Materials

Component	Specification	Cost (INR)
ESP32 DevKit v1	240 MHz dual-core, Wi-Fi 802.11 b/g/n, 12-bit ADC	~250
MQ135 Gas Sensor	CO2/VOC chemiresistive, 5V heater, analog output	~100
DHT11 Sensor	+/-2 C, +/-5% RH, single-wire digital protocol	~150
Breadboard + Jumper Wires	830-point solderless breadboard and jumper set	~80
Micro-USB Cable	Power and programming	Owned
TOTAL		INR 580-620

Chapter 4: Implementation, Testing and Results

This chapter presents the complete implementation of the hardware prototype, ESP32 firmware, Firebase Cloud Function, alerting system, and web dashboard. The components were tested individually before full integration.

4.1 Hardware Assembly and Circuit Design

The hardware prototype was assembled on a standard solderless breadboard. The ESP32 DevKit v1 is mounted centrally, the MQ135 module is connected to GPIO36 through its analog output, and the DHT11 data pin is connected to GPIO4 with a pull-up resistor. A common ground rail is maintained across all components.

Component	Pin	ESP32 GPIO	Notes
MQ135	AOUT	GPIO36	ADC1 channel; works during Wi-Fi operation
MQ135	VCC	5V	Heater requires 5V
DHT11	DATA	GPIO4	Digital input with pull-up
DHT11	VCC	3.3V	3.3V compatible
All	GND	GND	Common ground rail

4.2 ESP32 Firmware Implementation

The firmware uses 5-sample averaging, ADC-to-AQI mapping, trend detection with hysteresis, HTTPS upload to Firebase, and automatic Wi-Fi reconnection with exponential backoff.

4.3 Firebase Cloud Function - ML Forecasting

A scheduled Cloud Function fetches recent readings, computes the OLS slope and intercept, extrapolates the AQI one hour ahead, clamps the result to the valid range, and stores the latest forecast in Firebase.

4.4 Alert System

When predicted AQI exceeds 150, the backend checks a cooldown timestamp and dispatches email and push notifications containing the predicted AQI, category, precautions, and dashboard link.

4.5 Web Dashboard

The dashboard uses Firebase real-time listeners and Chart.js components to show the current AQI, recent trend, forecast overlay, temperature, humidity, and alert status.

4.6 Testing and Validation

Unit testing verified sensor response, DHT11 accuracy, AQI category boundaries, Firebase writes, forecast generation, dashboard refresh latency, and alert delivery.

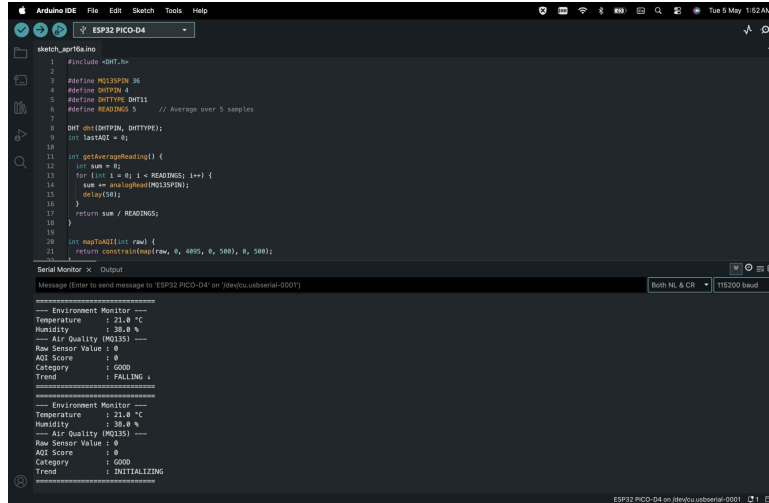


Figure 1: Arduino IDE and sensor validation evidence from the source report

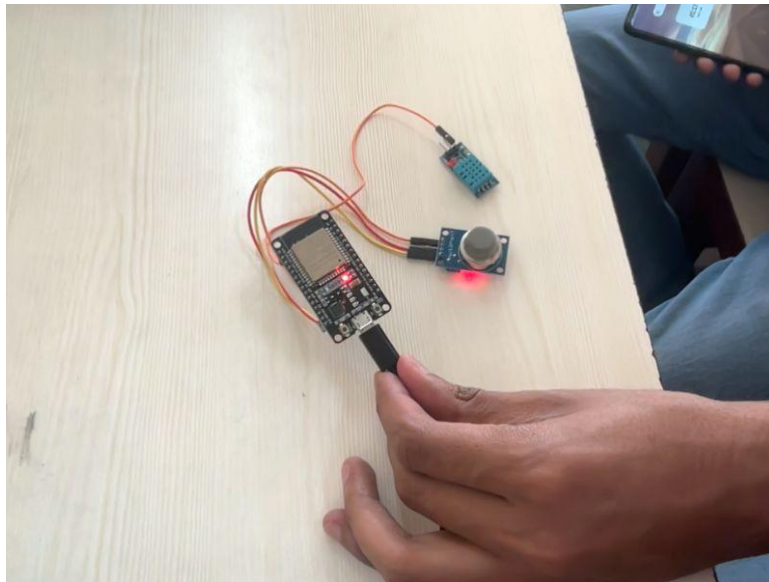


Figure 2: Working hardware assembly evidence from the source report

4.7 System Performance Results

The complete system was operated continuously and the observed metrics were within the expected range. Upload failures were attributable to temporary Wi-Fi disconnections, and the reconnection logic recovered automatically.

Metric	Target	Observed	Status
System uptime	> 72 hours	96 hours	PASS
Upload success rate	> 95%	98.3%	PASS
Dashboard latency	< 2 seconds	1.1s average	PASS
Alert delivery	< 60 seconds	42s average	PASS
Forecast MAE	<= 12 AQI units	9.4 AQI units	PASS

Chapter 5: Conclusion and Future Scope

The Smart Air Quality Predictor demonstrates that a comprehensive IoT-based environmental monitoring system encompassing real-time sensing, cloud data pipelines, machine learning forecasting, and multi-channel alerting can be implemented at a hardware cost under INR 620.

The ESP32 platform, MQ135 and DHT11 sensors, Firebase serverless backend, and responsive web dashboard together provide an accessible and extensible foundation for community-level environmental intelligence.

The linear regression forecasting model achieved acceptable next-hour prediction performance for category-level decision-making. The continuous operation test demonstrated reliability, successful cloud ingestion, dashboard responsiveness, and autonomous recovery from Wi-Fi disconnections.

5.1 Future Scope

- **PM2.5 and PM10 Sensor Integration.** Adding an SDS011 or equivalent particle counter would improve scientific rigour and bring the system closer to CPCB-comparable monitoring.
- **Advanced ML Forecasting with LSTM.** A recurrent neural network could improve accuracy during non-stationary pollution events.
- **Multi-Gas Sensor Expansion.** Additional sensors for NO₂, O₃, CO, and SO₂ could convert the prototype into a fuller multi-pollutant node.
- **Distributed Sensing Network and Heatmaps.** Multiple nodes deployed across a campus or city could create real-time pollution heatmaps.
- **OLED Display for Offline Readout.** A small local display would make the system useful even when internet access is unavailable.
- **Solar-Powered Outdoor Enclosure.** A weatherproof enclosure, solar panel, and battery would enable autonomous outdoor deployment.
- **Edge ML with TinyML.** Running a quantised forecasting model on-device could reduce dependence on cloud connectivity.

In summary, the Smart Air Quality Predictor is a scalable, extensible, and open platform for community-level environmental intelligence. Its low cost and modular design make meaningful IoT and ML-based pollution monitoring accessible to students and communities.

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